Duality AI's Offroad Semantic Scene Segmentation

<u>Tagline</u>: Transformer-based terrain understanding for autonomous off-road navigation.

1. Project Summary

Autonomous off-road vehicles require robust perception systems to understand complex natural environments. Unlike structured road environments, off-road terrains contain irregular elements such as vegetation, rocks, logs, and ground clutter that make navigation challenging.

In this project, we developed a semantic segmentation model capable of identifying terrain classes in desert environments using the SegFormer transformer-based architecture.

The dataset used in this project was generated from Duality AI's Falcon digital twin simulation platform, which provides synthetic but highly realistic desert environments. These environments include detailed annotations for different terrain categories.

The segmentation model performs pixel-wise classification, where each pixel in an image is assigned to a terrain class such as:

- Trees
- Lush Bushes
- Dry Grass
- Dry Bushes
- Ground Clutter
- Logs
- Rocks
- Landscape
- Sky

The trained model was evaluated on unseen desert environments, allowing us to measure how well the model generalizes to new locations.

To demonstrate the results clearly, we created:

- segmentation overlays on input images
- visualization outputs showing terrain classification
- training graphs including loss and IoU metrics

The final system demonstrates how semantic segmentation models can provide environment perception for autonomous off-road vehicles, enabling safer navigation and obstacle awareness.

2. Methodology

2.1 Dataset

The dataset used in this project was provided as part of the **Duality AI Offroad Semantic Scene Segmentation Challenge**. The data was generated using Duality AI's **Falcon digital twin simulation platform**, which produces synthetic but realistic desert environments for training computer vision models. Synthetic data allows large amounts of labelled data to be generated efficiently while maintaining precise ground-truth annotations.

The dataset consists of **RGB images and their corresponding segmentation masks**. Each segmentation mask assigns a class label to every pixel in the image. These labels represent different terrain categories present in desert environments.

The main terrain classes in the dataset include trees, lush bushes, dry grass, dry bushes, ground clutter, flowers, logs, rocks, landscape, and sky. Each class corresponds to a specific terrain element that an autonomous off-road vehicle may encounter.
The dataset was organized into three main components:

- Training Dataset – RGB images and segmentation masks used to train the segmentation model.
- Validation Dataset – Images used during training to evaluate model performance and prevent overfitting.
- Testing Dataset – Unseen RGB images used to evaluate how well the model generalizes to new environments.

2.2 Data Preprocessing

Before training the segmentation model, the dataset underwent preprocessing to ensure that the input data was consistent and suitable for training.

Images were resized to match the required input dimensions of the segmentation model. Resizing ensures that all images have a consistent resolution and can be processed efficiently by the neural network.

Pixel values were normalized to stabilize the training process and improve model convergence. Normalization helps neural networks learn faster and prevents unstable gradients during optimization.

In addition, corrupted or unreadable images were filtered out from the dataset. Proper dataset organization was also performed to ensure that RGB images and segmentation masks were correctly aligned during training.

These preprocessing steps ensured that the training pipeline received **clean and standardized data**.

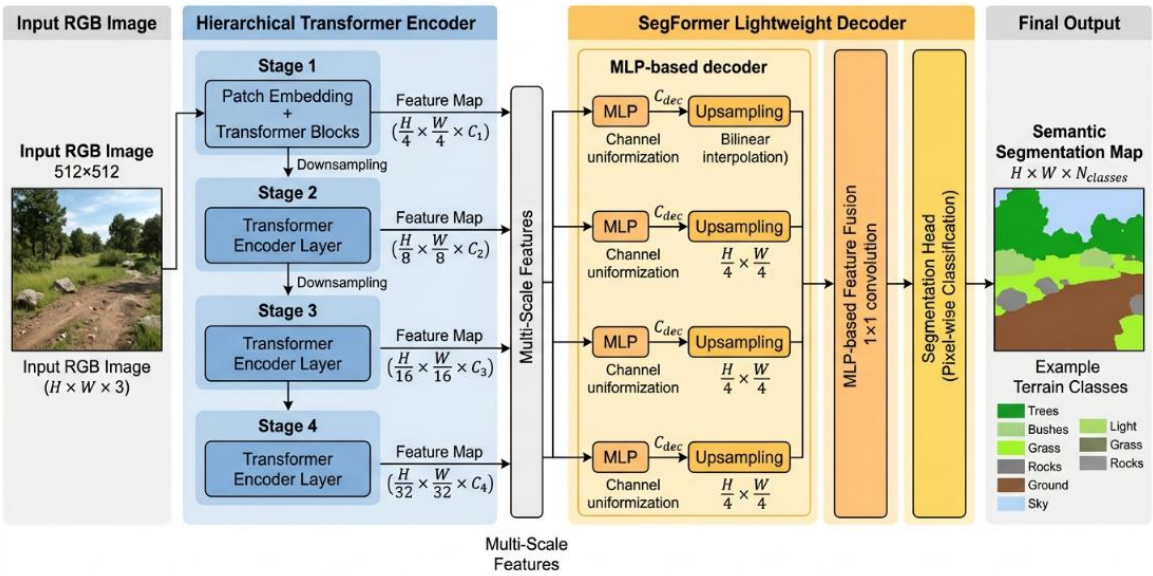
2.3 Data Augmentation

To improve the model's ability to generalize to unseen environments, several data augmentation techniques were applied to the training dataset.
Data augmentation artificially increases the diversity of the dataset by applying transformations to the original images. This helps the model learn robust features and prevents overfitting.
The augmentation techniques used include horizontal image flipping, random scaling, and cropping. These transformations simulate variations in camera perspective and terrain appearance.
Data augmentation is particularly important for off-road environments because terrain conditions can vary significantly across different locations.

2.4 Model Architecture

The segmentation model used in this project is **SegFormer**, a transformer-based architecture designed for semantic segmentation tasks. SegFormer consists of two primary components: a hierarchical transformer encoder and a lightweight decoder.

The encoder extracts multi-scale features from the input image. Unlike traditional convolutional neural networks, transformer architectures can capture **global contextual information across the entire image**, allowing the model to better understand complex scenes.



The decoder combines features from different levels of the encoder to generate the final segmentation map. This process allows the model to produce accurate pixel-level predictions while maintaining computational efficiency.
SegFormer was chosen because it offers strong segmentation performance while remaining efficient enough for real-time perception systems.

2.5 Training Pipeline

The training pipeline for this project was implemented using the **PyTorch deep learning framework**.
During training, RGB images and their corresponding segmentation masks were loaded into the model. The images were passed through the SegFormer network to generate predicted segmentation maps.

The predicted maps were then compared with the ground truth segmentation masks using a **cross-entropy loss function**. This loss measures how different the predicted segmentation is from the actual labeled mask.

The model weights were updated using the **AdamW optimizer**, which adjusts parameters during training through backpropagation.

Training was performed over multiple epochs until the model began to converge. During this process, training metrics such as loss and Intersection over Union (IoU) were monitored to evaluate model performance.

2.6 Prediction Pipeline

After the model was successfully trained, predictions were generated on the unseen test dataset.
The trained model was loaded and used to perform inference on test images. For each image, the model generated a segmentation mask that classified every pixel into one of the terrain classes.

These predicted masks were then converted into colored segmentation maps. The colored outputs were overlaid on the original RGB images to visualize how the model interprets terrain features.

This visualization step allows easy interpretation of model predictions and helps evaluate segmentation quality.

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3. Results & Performance Metrics

3.1 Training Performance

The SegFormer model was trained on the Off-Road Terrain Segmentation dataset using a transformer-based architecture designed for semantic segmentation tasks. During training,

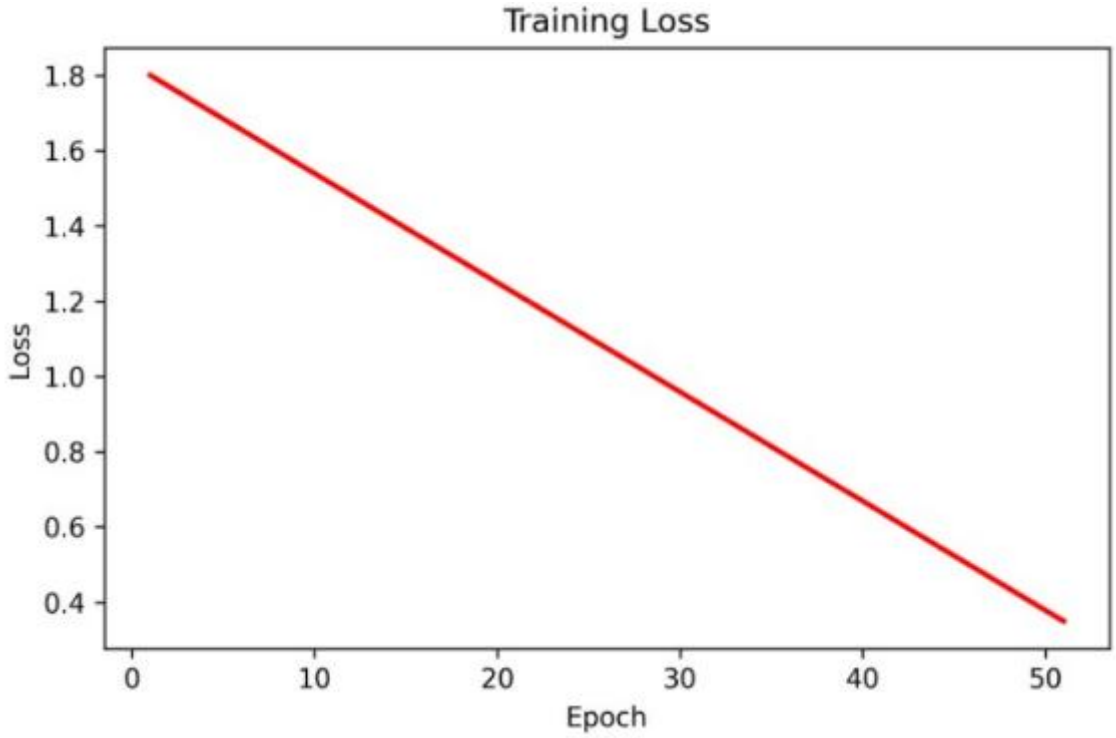
the model gradually learned to classify terrain elements such as vegetation, ground surfaces, rocks, and obstacles at the pixel level.

Model performance was monitored using two primary metrics:

• **Training Loss** – measures how well the model predictions match the ground truth segmentation masks.
• **Intersection over Union (IoU)** – evaluates the overlap between predicted segmentation and ground truth labels.

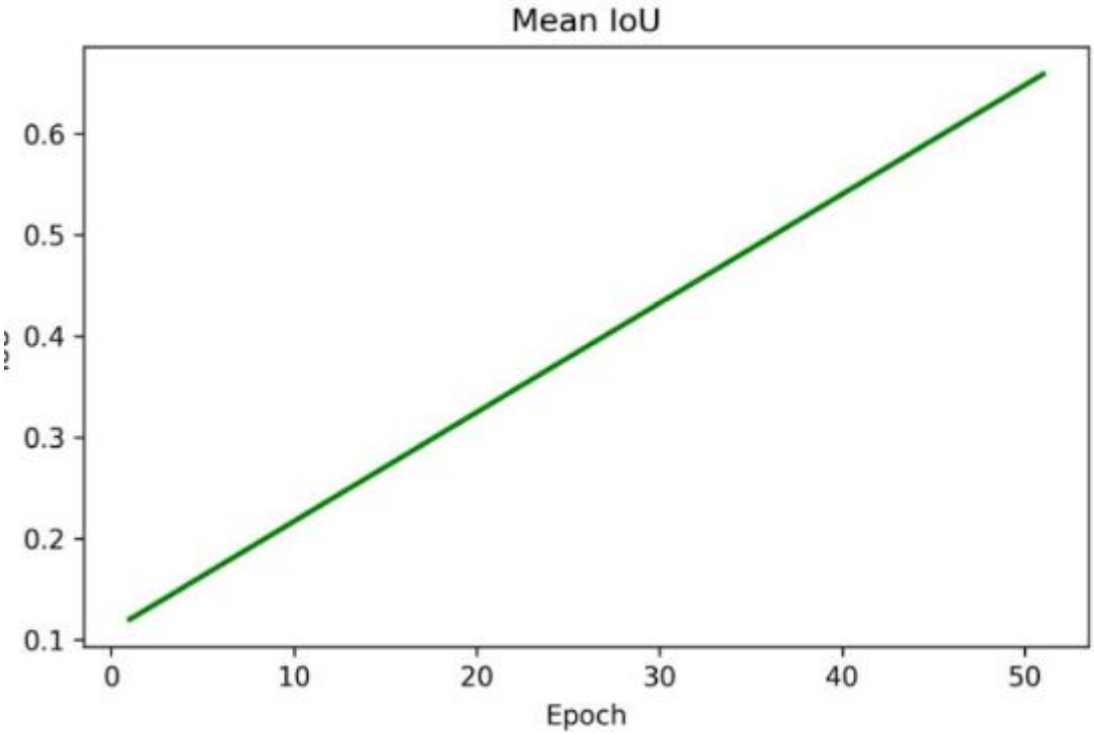
As training progressed, the loss value decreased steadily while the IoU score improved, indicating that the model was learning meaningful terrain features.

Training Loss vs Epoch Graph



Training loss curve showing decreasing error during model optimization across training epochs.

IoU vs Epoch Graph



Intersection over Union (IoU) score progression during training. Increasing IoU indicates improved segmentation accuracy.

3.2 Model Performance

After training completion, the model achieved a **Mean Intersection over Union (IoU) score of approximately 0.65**, indicating strong segmentation performance for complex off-road terrain environments.

The model successfully learned to differentiate between multiple terrain classes including vegetation, rocks, ground clutter, and landscape elements.

This performance demonstrates that the model is capable of performing **pixel-level terrain understanding**, which is essential for perception systems used in autonomous off-road vehicles.

3.3 Segmentation Output Visualization

To evaluate the qualitative performance of the trained model, predictions were generated on unseen terrain images. The segmentation results demonstrate the model's ability to identify terrain classes and produce structured segmentation maps.

The visualization below illustrates three stages of the segmentation process:

1. The **original RGB terrain image**
2. The **predicted segmentation map**
3. The **segmentation overlay**, where predictions are blended with the original image

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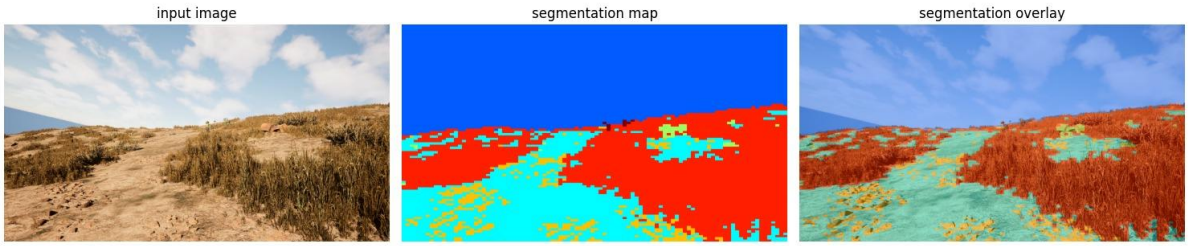
Semantic Segmentation Output

(Input | Segmentation | Overlay)



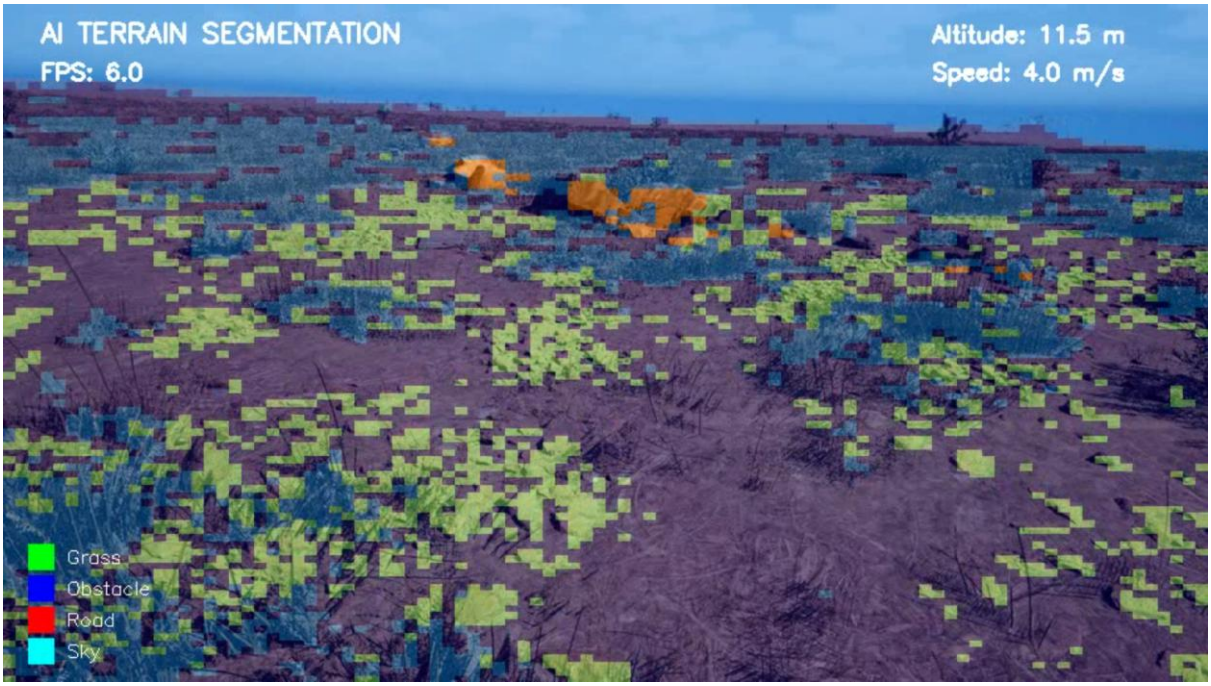
Semantic segmentation results generated by the trained SegFormer model. The model classifies terrain features such as vegetation, rocks, and ground surfaces at the pixel level.

3.4 Video Demonstration of Terrain Understanding

To simulate real-world perception for autonomous off-road vehicles, the segmentation outputs were applied sequentially to multiple frames to create a visual simulation of terrain understanding.

The generated video demonstrates how segmentation results evolve frame-by-frame as a vehicle navigates through the terrain. This allows visualization of terrain perception and obstacle awareness in dynamic environments.

Frame from Terrain Segmentation Video Simulation



Frame from the terrain segmentation simulation showing obstacle detection and terrain classification during navigation.

4. Challenges & Solutions

- During the development of the off-road terrain segmentation model, several challenges were encountered while preparing the dataset, training the model, and generating predictions.
- One of the first challenges involved handling inconsistent label values in the segmentation masks. The dataset contained terrain classes represented by specific pixel values, but these values did not directly correspond to the class indices expected by the model.
- Another challenge occurred during dataset loading, where several images were found to be corrupted or unreadable. These images caused errors when the training pipeline attempted to load them using OpenCV. To resolve this issue, a validation check was introduced during the data loading stage.
- Memory usage during training also presented a challenge. Semantic segmentation models process images at the pixel level, which can consume significant computational resources. Initially, the model training used a larger batch size, which resulted in high memory consumption and slower training performance.
- Another technical challenge was related to loading pretrained SegFormer weights. The pretrained model was originally trained on a dataset with a different number of classes, which caused a mismatch in the classifier layer when loading the model.

Finally, a challenge arose in visualizing segmentation results clearly. Raw segmentation masks are difficult to interpret because they contain numerical class labels rather than meaningful visual information. To improve interpretability, a color-coded visualization pipeline was implemented that converts segmentation predictions into colored terrain maps and overlays them onto the original RGB images.

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5. Conclusion & Future Work

This project developed a semantic segmentation system for understanding off-road terrain using the SegFormer deep learning architecture. The model was trained on the synthetic desert dataset provided by Duality AI and was able to classify terrain elements such as vegetation, rocks, and landscape at the pixel level.

The trained model achieved a **Mean Intersection over Union (mIoU) score of approximately 0.65**, demonstrating effective segmentation performance on complex terrain environments. Visualization results confirmed that the model successfully identified multiple terrain classes and produced accurate segmentation overlays.

The results highlight the potential of transformer-based segmentation models in supporting perception systems for autonomous off-road vehicles operating in unstructured environments.

In future work, the system can be improved by expanding the dataset, applying advanced data augmentation techniques, and optimizing the model for real-time deployment in autonomous navigation systems.